Module-5

Definition and examples of linear transformations. Kernel and range, Matrix representation of a linear transformation, Change of basis. Similarity of matrices, Invariant subspaces, Minimal polynomial of a linear operator.

LINEAR TRANSFORMATIONS

Definition of Linear Transformation

A function $T: V \to W$ between two vector spaces is called a **linear transformation** if for all vectors $\mathbf{u}, \mathbf{v} \in V$ and scalars c:

1. **Additivity:**

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$$

2. **Homogeneity (scalar multiplication):**

$$T(c\mathbf{u}) = c\,T(\mathbf{u})$$

If both conditions hold, T is linear.

Problem 1

Let $T: \mathbb{R}^2 \to \mathbb{R}$ be,

$$T(x, y) = 3x - 4y$$

Find $T(2, -1)$.

Solution

$$T(2, -1) = 3(2) - 4(-1) = 6 + 4 = 10$$

Problem 2. Let $T: P_2 \to P_1$ be defined by

$$T(a + bx + cx^2) = b + 2cx.$$

Find:

1. $T(1 + 2x - x^2)$
2. $T(3 - x + 4x^2)$

Solution

(1)

$$a = 1,\ b = 2,\ c = -1$$

$$T(1 + 2x - x^2) = 2 + 2(-1)x = 2 - 2x$$

(2)

$$a = 3,\ b = -1,\ c = 4$$

$$T(3 - x + 4x^2) = -1 + 8x$$

Problem 3

Let $\boldsymbol{T}: \mathbb{R}^3 \to \mathbb{R}^2$be

$$\boldsymbol{T(x, y, z) = (x - y, y + z)}$$

Find $\boldsymbol{T(1, 2, 3)}$and $\boldsymbol{T(-1, 0, 4)}$.
Solution

$$\boldsymbol{T(1, 2, 3) = (1 - 2, 2 + 3) = (-1, 5)}$$
$$\boldsymbol{T(-1, 0, 4) = (-1 - 0, 0 + 4) = (-1, 4)}$$

Kernel and Range

Definition of Kernel and Range
Let

$$T: V \to W$$

be a linear transformation between vector spaces.

Kernel of a Linear Transformation

The **kernel** of $T$is the set of all vectors in Vthat are mapped to the zero vector in W.

$$\ker(T) = \{ v \in V \mid T(v) = 0 \}$$

It is always a subspace of the domain.

Range of a Linear Transformation

The **range** (or image) of $T$is the set of all vectors in Wthat can be written as $T(v)$for some $v \in V$.

$$\text{Range}(T) = \{ T(v) \mid v \in V \}$$

It is always a subspace of the codomain.
Problem 1
Let

$$T: \mathbb{R}^3 \to \mathbb{R}^2, T(x, y, z) = (x + y,\ y + z)$$

Find:
1. $\ker(T)$
2. $\text{Range}(T)$

Solution
Step 1: Find the Kernel
Solve

$$T(x, y, z) = (0,0)$$

This gives the system:

$$x + y = 0$$
$$y + z = 0$$

From the first equation → $x = -y$
From the second → $z = -y$
Let $y = t$.
So,

$$(x, y, z) = (-t, t, -t) = t(-1,1,-1)$$

Kernel

$$\ker(T) = \text{span}\{(-1,1,-1)\}$$

Step 2: Find the Range
A linear transformation from matrix:

$$T(x,y,z) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

The column vectors are:

$$c_1 = (1,0), c_2 = (1,1), c_3 = (0,1)$$

The range = span of columns:

$$\text{Range}(T) = \text{span}\,\{(1,0), (1,1), (0,1)\}$$

Problem 2

Let

$$T: \mathbb{R}^2 \to \mathbb{R}^2, T(x,y) = (2x, 3y)$$

Find **Kernel** and **Range**.

Solution

Kernel

Solve

$T(x,y) = (0,0) \Rightarrow (2x, 3y) = (0,0)$ *and* $x = 0 = y$

Range

Outputs are of the form:

$$(2x, 3y)$$

Thus

$$\text{Range}\,(T) = \{(u,v): u = 2x, v = 3y\} = \mathbb{R}^2$$

Problem 3

Let

$$T(x,y) = (x+y, x-y)$$

Find Kernel and Range.

Solution

Kernel

$$x + y = 0, x - y = 0$$

Add equations:

$$2x = 0 \Rightarrow x = 0 \Rightarrow y = 0$$
$$\ker\,(T) = \{(0,0)\}$$

Range

Matrix:

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Determinant:

det(A)=−2

Problem 4

Let

$$T(x,y) = (3x - 6y,\ x + 2y)$$

Solution

Solve

$$3x - 6y = 0, x + 2y = 0$$

From 2nd equation:

$$x = -2y$$

Substitute in 1st equation:

$$3(-2y) - 6y = -6y - 6y = -12y = 0 \Rightarrow y = 0 \Rightarrow x = 0$$

Kernel

$$ker\ (T) = \{(0,0)\}$$

Problem 5

Let

$$T(x, y, z) = (x - y, y - z)$$

Find range.

Solution

Matrix:

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

Rank = 2 (rows independent).

Thus

$$\text{Range}\ (T) = \mathbb{R}^2$$

Problem 6

Let

$$T(x, y, z) = (2x - 4y + z)$$

Find **kernel** and **range**.

Solution

This is mapping to $\mathbb{R}$.

Kernel:

$$2x - 4y + z = 0$$

Let $y = s, z = t$:

$$2x = 4s - t \Rightarrow x = 2s - t/2$$

Thus kernel:

$$\ker(T) = \{(2s - t/2,\ s,\ t) : s, t \in \mathbb{R}\}$$

Range: Since output is one equation with real values →

$$\text{Range}(T) = \mathbb{R}$$

Problem 7

Let

$$T(x, y, z) = (x + y + z, x + 2y + 3z, x + 3y + 6z)$$

Find kernel and range.

Solution

Matrix:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{pmatrix}$$

Row reduce:
R2 = R2 − R1
R3 = R3 − R1

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 5 \end{pmatrix}$$

R3 = R3 − 2R2

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

Rank = 3 → full rank.
Kernel

$$\ker(T) = \{(0,0,0)\}$$

Range

$$\text{Range}(T) = \mathbb{R}^3$$

Matrix representation of a linear transformation

If $T: V \to W$ is a linear transformation and
$\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ is a basis of V,
$\mathcal{C} = \{w_1, w_2, \dots, w_m\}$ is a basis of W,
The matrix representation of T with respect to $\mathcal{B}$ and $\mathcal{C}$ is:

$$[T]_{\mathcal{C}}^{\mathcal{B}} = [[T(v_1)]_{\mathcal{C}} \quad [T(v_2)]_{\mathcal{C}} \quad \cdots \quad [T(v_n)]_{\mathcal{C}}]$$

Problem 1

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$T(x, y) = (2x + y,\ x - y).$$

Find the standard matrix representation of T.
Solution
Use standard basis vectors:
Step 1: Compute $T(e_1)$

$$e_1 = (1,0), T(1,0) = (2 \cdot 1 + 0,\ 1 - 0) = (2,1)$$

Step 2: Compute $T(e_2)$

$$e_2 = (0,1), T(0,1) = (0 + 1,\ 0 - 1) = (1, -1)$$

Step 3: Form matrix
Columns are $T(e_1)$ and $T(e_2)$:

$$[T] = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

Problem 2
Let $T: \mathbb{R}^3 \to \mathbb{R}^2$ be given by

$$T(x, y, z) = (x + 2y - z,\ 3x - y + 4z).$$

Find the standard matrix representation of T.
Solution
Compute outputs of basis vectors:

$$\boldsymbol{T(e_1)}$$
$$T(1,0,0) = (1,3)$$
$$T(0,1,0) = (2,-1)$$
$$\boldsymbol{T(e_3)}$$
$$T(0,0,1) = (-1,4)$$

Form the matrix:

$$[T] = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{bmatrix}$$

Problem 3
Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be given by

$$T(x, y) = (3x - y,\ x + 4y)$$

Let basis

$$\mathcal{B} = \{(1,1), (1,-1)\}$$

Find the matrix of T w.r.t. basis $\mathcal{B}$.

Solution

Step 1: Apply T on basis vectors

1. $v_1 = (1,1)$

$$T(1,1) = (3 - 1,\ 1 + 4) = (2,5)$$

2. $v_2 = (1,-1)$

$$T(1,-1) = (3 + 1,\ 1 - 4) = (4,-3)$$

Step 2: Express results in basis $\mathcal{B}$

We solve:

$$(2,5) = a(1,1) + b(1,-1)$$
$$(2,5) = (a + b,\ a - b)$$

Solve system:

$$a + b = 2$$

Add:

$$2a = 7 \Rightarrow a = 3.5$$

$$b = -1.5$$

So:

$$[T(v_1)]_{\mathcal{B}} = \begin{bmatrix} 3.5 \\ -1.5 \end{bmatrix}$$

Similarly for $(4, -3)$:

$$(4, -3) = a(1,1) + b(1, -1)$$

Solve:

$$\begin{aligned} a + b &= 4 \\ a - b &= -3 \end{aligned}$$

Add:

$$\begin{aligned} &2a = 1 \Rightarrow a = 0.5 \\ &b = 3.5 \end{aligned}$$

Thus:

$$[T(v_2)]_{\mathcal{B}} = \begin{bmatrix} 0.5 \\ 3.5 \end{bmatrix}$$

Final Matrix

$$[T]_{\mathcal{B}} = \begin{bmatrix} 3.5 & 0.5 \\ -1.5 & 3.5 \end{bmatrix}$$

Problem 4

Let

$$T: \mathbb{R}^2 \to \mathbb{R}^2, T(x, y) = (4x + y,\ 2x + 3y)$$

Given bases:

$$\mathcal{B} = \{(1,2), (2,1)\}, \mathcal{C} = \{(1,1), (1, -1)\}$$

Find the matrix representation $[T]_{\mathcal{C}}^{\mathcal{B}}$.

Solution

Step 1 — Compute $T(v_1)$and $T(v_2)$

For $v_1 = (1,2)$:

$$T(1,2) = (4 + 2,\ 2 + 6) = (6,8)$$

For $v_2 = (2,1)$:

$$T(2,1) = (8 + 1,\ 4 + 3) = (9,7)$$

Step 2 — Express in basis $\mathcal{C} = \{(\mathbf{1}, \mathbf{1}), (\mathbf{1}, -\mathbf{1})\}$
Solve for $(6, 8)$:

$$(6,8) = a(1,1) + b(1,-1)$$

Equations:

$$a + b = 6$$
$$a - b = 8$$

Add:

$$2a = 14 \Rightarrow a = 7$$
$$b = -1$$

Thus:

$$[T(v_1)]_{\mathcal{C}} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

For $(9, 7)$:

$$a + b = 9$$
$$a - b = 7$$

Add:

$$2a = 16 \Rightarrow a = 8$$
$$b = 1$$

Thus:

$$[T(v_2)]_{\mathcal{C}} = \begin{bmatrix} 8 \\ 1 \end{bmatrix}$$

Final Matrix

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{bmatrix} 7 & 8 \\ -1 & 1 \end{bmatrix}$$

Similarity of matrices

Two matrices $A$and $B$are similar if there exists an invertible matrix Psuch that:

$$\boldsymbol{B = P^{-1}AP}$$

Similarity means:

- They represent the same linear transformation under different bases.
- They have the same eigenvalues, determinant, rank, trace, and characteristic polynomial.

Problem 1

Determine whether the matrices

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 3 & 2 \end{bmatrix}$$

are similar.

Solution

Similarity requires same characteristic polynomial.

Step 1: Characteristic polynomials

For A:

$$\chi_A(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix} = (2-\lambda)^2$$

For B:

$$\chi_B(\lambda) = \det \begin{bmatrix} 2-\lambda & 0 \\ 3 & 2-\lambda \end{bmatrix} = (2-\lambda)^2$$

Step 2: Check minimal polynomials

Both are non-diagonal but have one Jordan block.
Thus they have:

- Same characteristic polynomial ✓
- Same minimal polynomial ✓

✓ Conclusion:

$$A \sim B$$

Problem 2
Let

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

Find a matrix Psuch that

$$P^{-1}AP = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

if possible.
Solution
Step 1: Check diagonalizability
Eigenvalue: $\lambda = 3$(repeated twice).
Compute eigenvectors:

$$A - 3I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Solving:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow y = 0$$

Eigenvector = $(1, 0)$. Only one eigenvector ⇒ not diagonalizable.
Conclusion:
No matrix Pexists because $A$is not diagonalizable.

Problem 3
Check whether

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

are similar.

Solution

Similarity requires same trace and determinant.

Compute trace:

$$\text{tr}\,(A) = 1 + 4 = 5$$
$$\text{tr}\,(B) = 5 + 8 = 13$$

Traces are NOT equal.

Conclusion:

$$A \sim / B$$

Problem 4
Show that

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \text{and} B = \begin{bmatrix} 2 & 3 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

are similar.
Solution
Both are upper triangular with diagonal entries:

$$2,2,2$$

Same eigenvalues (all 2).
Both are Jordan block matrices of size 3.
So they represent the same Jordan form:

$$J = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Thus each is similar to J, hence:

$$A \sim J \sim B \Rightarrow A \sim B$$